

KHAIRUNNISA KHAN BINTI TALIB

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SUMMARY

GenAI engineer who builds LLM-powered products end to end: multi-agent systems in Python (LangGraph, RAG, evaluation) and the React/Next.js/TypeScript apps around them. Machine-learning foundation from the University of Manchester (BSc Computer Science). Re-architected a production learning platform into a multi-agent service with task decomposition, retrieval-grounded answers, and CI-run LLM evaluations; built a government digital-platform prototype used in a national tender; shipped CI-integrated Playwright suites. AWS Certified Solutions Architect - Associate.

TECHNICAL SKILLS

- **GenAI / Agents:** LangGraph, multi-agent orchestration, RAG (chunking, embeddings, vector retrieval), LLM evaluation & LangSmith tracing, Pydantic structured output, Groq / Llama, Model Context Protocol (MCP), Hugging Face Transformers
- **Machine Learning:** PyTorch, TensorFlow, Keras, scikit-learn, BERT, Bi-LSTM, NLP
- **Languages:** Python, JavaScript, TypeScript, Java, SQL
- **Backend & Infra:** FastAPI, Flask, Spring Boot, Docker, GitHub Actions CI/CD, AWS (certified), Azure Container Apps, Vercel, n8n
- **Frontend:** React, Next.js (App Router), Tailwind CSS, Material UI (Material 3), Recharts, Framer Motion
- **Testing & Quality:** Playwright (UI, API, accessibility, regression), Test-Driven Development, LLM eval harnesses
- **Data & Databases:** MySQL, MongoDB, Oracle SQL, Power BI, data cleaning & visualization

WORK EXPERIENCE

Frontend Developer | Artech Niaga Sdn Bhd

Oct 2025 - Present

- Built a functional prototype of the MAQIS Digital Platform, a national quarantine and inspection management system, using Next.js, React, TypeScript, and Tailwind CSS; deployed on Vercel for stakeholder evaluation.
- Developed the admin portal with authentication flows, bilingual (Malay/English) UI, and interactive Recharts dashboards for operational reporting.
- Authored technical documentation and compliance materials that supported the company's government tender submission.

Solution Engineering Intern | Infobip Solution Malaysia

Mar 2025 - Jun 2025

- Supported client engagement sessions, including live product demos, proof-of-concept presentations, and technical Q&A for enterprise messaging solutions.
- Customized chatbot implementations to customer requirements, translating business needs into working conversational flows.
- Prepared User Acceptance Testing (UAT) documentation to support client-side solution validation and sign-off.

PROJECTS

Bangkit - Multi-Agent AI Learning Platform

Python, LangGraph, FastAPI, RAG, Next.js, Docker

- Re-architected the AI layer of a production SPM learning platform into a multi-agent LangGraph service (Python/FastAPI) that performs task decomposition, planning, and execution across tutor, quiz-generation, and marking agents coordinated by a supervisor.
- Built a RAG pipeline over curriculum content and digitized SPM past papers (markdown-aware chunking, BGE embeddings, metadata-filtered vector store) so tutor answers stay grounded and cite the exact chapter and section.
- Enforced schema-validated structured output (Pydantic) with a self-healing repair step for quiz generation and marking, and a resilient multi-model LLM router with exponential-backoff fallback across Groq-hosted models (Llama 3.3 70B).
- Created an LLM evaluation harness with golden datasets and LangSmith tracing, measuring marking accuracy and answer groundedness in CI; containerized with Docker and a GitHub Actions pipeline deploying to Azure Container Apps / AWS App Runner.

SetupHub - E-commerce Test Automation Suite

Playwright, Node.js, Express, GitHub Actions

- Built an end-to-end Playwright automation suite for a Node/Express e-commerce application covering UI flows, API validation, regression scenarios, accessibility checks, and mocked network-failure conditions.
- Integrated the suite into GitHub Actions CI with Playwright HTML reporting for repeatable, reviewable test runs on every push.

Evidence Detection with NLP

Python, PyTorch, BERT, Bi-LSTM

- Trained and evaluated Bi-LSTM and BERT models for evidence detection, achieving 90% accuracy with both architectures and comparing recurrent versus transformer trade-offs.

EDUCATION

BSc (Hons) Computer Science - 2:1 (67.5%) | University of Manchester, UK

2021 - 2024

- Relevant coursework: Natural Language Processing, Machine Learning, Database Design (Oracle, SQL).

CERTIFICATIONS

- AWS Certified Solutions Architect - Associate (2025-2028); AWS Certified Cloud Practitioner; CompTIA Cloud Essentials+; Google Data Analytics Professional Certificate
- Test-Driven Development for JavaScript (Pearson/Coursera); Introduction to Model Context Protocol (Anthropic); Workflow Automation with n8n (Coursera)

ADDITIONAL

- Languages: English (IELTS Band 7.5), Malay (native), Arabic (basic). Public speaking: 2nd runner-up, Forum Remaja national final; national-level parliamentary-style debate participant.